

EDUCATION

Georgia State University

Master of Actuarial Science

Atlanta, GA, USA

August 2027 – December 2028 (*Expected*)

National Yang Ming Chiao Tung University

Bachelor of Science in Applied Mathematics

Hsinchu City, Taiwan

August 2022 – June 2026

École polytechnique

International Exchange Program

Palaiseau, France

September 2024 – June 2025

SKILLS

Programming Python, C++, and R; $\text{\LaTeX}$ and **typst**.

Languages Mandarin, English (CEFR C1), Japanese (JLPT N2), and French (conversational).

ASA Courses Exam P (July 2025), Exam FM (December 2025), VEE Mathematical Statistics, and VEE Accounting & Finance.

PROJECTS

Motor Insurance Pure Premium Pricing: Bridging Machine Learning and Actuarial Compliance using Tree-GLM Hybrids

March 2025 – Present

Independent Research Project

Submitted to NAAJ and under review.

- Developed a three-stage pipeline of rule extraction, binning, and penalized GLM to evaluate pure premiums using the `freMTPL2` dataset.
- Verified that the additive structure of Tree-GLM imposes no statistically detectable risk-ranking penalty relative to LightGBM on the `freMTPL2` dataset while preserving inspectability.
- Showcased that Tree-GLM's risk-ranking is dominated by behavioral (`BonusMalus`) rather than demographic covariates (`Region`).

Study of Hitting Time and Probability for Random Walks in One and Two Dimensions¹

July 2025 – February 2026

Research Project, College Student Research Scholarship, [National Science and Technology Council](#)

- Derived hitting time distributions and expectations for one-dimensional and two-dimensional simple random walks.
- Established an N^2 scaling for expected hitting times in bounded domains and linked discrete probability coefficients to torsional rigidity in beam theory, providing a cross-disciplinary analytical framework for evaluating boundary behaviors in random walks.

The Gambler's Ruin Problem for One-Dimensional Random Walks: Simple Symmetric Random Walk and Some Extensions¹

September 2025 – December 2025

Semester Project, [Individual Directed Study \(I\)](#), National Yang Ming Chiao Tung University

- Verified the solution x/r for simple symmetric walks and extended the scaling $\asymp (x+1)/r$ to finite-step-length and finite-variance models, adapting classical stochastic models to broader conditions.
- Awarded **Best Project** (Departmental) and **Best Presentation** (2025 Inter-University Conference).

Credit Card Default Detection Using Machine Learning

February 2025 – June 2025

Final Group Project, [Machine Learning](#), École polytechnique

- Engineered predictive pipelines in Python (`scikit-learn`) to address imbalanced datasets via optimizing the F1-score to capture rare default events.

¹Supervised by Prof. [Yuki CHINO](#).

- Evaluated the trade-off between interpretability and accuracy by comparing baseline logistic regression against ensemble methods like XGBoost, demonstrating that transparent models can achieve comparable performance.

Chessbot Project: Cagnus

November 2024 – January 2025

Final Group Project, [Object-oriented Programming](#), École polytechnique

- Implemented bitboard representations and minimax evaluation in pure C++.
- Placed 9th of 26 in the class competitive tournament.

Before Finance¹

June 2024 – July 2024

[Summer Research](#), Department of Applied Mathematics, National Yang Ming Chiao Tung University

- Analyzed hitting times for simple random walks in *hitting time of 1* and the *gambler's ruin problem*.
- Constructed measure-theoretic proofs for martingale properties, including the optional sampling theorem, establishing a theoretical foundation for stochastic analysis and quantitative modeling.

WORK EXPERIENCE

Teaching Assistant

September 2025 – June 2026 & September 2023 – June 2024

Department of Applied Mathematics, National Yang Ming Chiao Tung University

AY 2025 [Probability Theory](#) and [Calculus A \(II\)](#) under Prof. [FANG Xiang](#).

AY 2023 [Calculus A \(I\)](#) and [Calculus A \(II\)](#) under Prof. Yuki CHINO.

Private Tutor

March 2026 – June 2026, September 2024 – June 2025, & April 2023 – June 2023

Tutored students with customized handouts.

Academic Tutor

October 2023 – November 2023 & April 2024 – May 2024

Center of Teaching and Learning Development, National Yang Ming Chiao Tung University

Tutored overseas students in Calculus A and conducted weekly sessions with customized [handouts](#).

HONORS

Mathematics Presidential Award

March 2026, September 2023, & February 2023

Department of Applied Mathematics, National Yang Ming Chiao Tung University

For my academic performance in applied mathematics in my undergraduate study.

College Student Research Scholarship

July 2025 – February 2026

National Science and Technology Council

For the project “Study of Hitting Time and Probability for Random Walks in One and Two Dimensions” under research grant 114-2813-C-A49-139-M.

PTSGL.com 23rd Translation Scholarship

January 2026

[PTSGL.com](#) (Presidential Translation Service Group International)

Selected as one of the 30 recipients in the translation competition for Japanese-to-English translation.

College of Science Dean Award

October 2025

College of Science, National Yang Ming Chiao Tung University

For ranking in the top 10% of the department in my first three years of undergraduate study.

Outstanding Student Overseas Exchange Scholarship

June 2024

National Yang Ming Chiao Tung University & the Ministry of Education

For my exchange program at École polytechnique in France.

Honor Student

May 2023

National Yang Ming Chiao Tung University

For ranking in the top 5% of the department in my first semester of undergraduate study.